

Paper 1 - Set 03 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	A	Statistics
2	C	Statistics
3	B	Polynomials, factor and remainder theorems
4	D	Indices and surds
5	A	Graphs of functions
6	A	Straight lines
7	E	Integration
8	E	Trigonometric functions and equations
9	D	Integration
10	A	Sequences and series
11	B	Binomial expansion
12	E	Differentiation
13	E	Indices and surds
14	C	GCSE algebra
15	A	Probability
16	D	Binomial expansion
17	D	Functions and transformations
18	A	Simultaneous equations and inequalities
19	A	Triangles, radians and sectors
20	B	Differentiation

Worked solutions

Question 1 STA-01 · Statistics · spec M6.3

The mean of five numbers is 12. When a sixth number is added the mean becomes 14. What is the sixth number?

Answer: A

Total goes from 60 to 84: the sixth number is 24.

Question 2 STA-03 · Statistics · spec M6.3

The numbers 3, 7, x , 12, 18 (with $7 \leq x \leq 12$) have mean equal to median. Find x .

Answer: C

The median is x and the mean is $\frac{40+x}{5}$. Setting them equal: $40 + x = 5x$, so $x = 10$.

Question 3 POL-03 · Polynomials, factor and remainder theorems · spec MM1.6

Fully factorise $x^3 - 7x + 6$.

Answer: B

$f(1) = 0$, so $(x - 1)$ is a factor: $x^3 - 7x + 6 = (x - 1)(x^2 + x - 6) = (x - 1)(x + 3)(x - 2)$.

Question 4 IND-20 · Indices and surds · spec MM1.2

Given that $\sqrt{a} - \sqrt{b} = 2$ and $a - b = 12$, find $\sqrt{a} + \sqrt{b}$.

Answer: D

$a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})$, so $12 = 2(\sqrt{a} + \sqrt{b})$, giving 6.

Question 5 GRA-06 · Graphs of functions · spec MM8.6

The curve $y = x^4 - 5x^2 + 4$ meets the coordinate axes at how many points in total?

Answer: A

x -axis: $(x^2 - 1)(x^2 - 4) = 0$, four points. y -axis: $(0, 4)$, one point. Total 5.

Question 6 LIN-05 · Straight lines · spec MM3.1

Which of the following lines is parallel to $4x + 6y = 9$?

Answer: A

$4x + 6y = 9$ has gradient $-\frac{2}{3}$; so does $2x + 3y = 1$.

Question 7 INT-08 · Integration · spec MM7.4

Let $F(x) = \int_2^x (t^2 - 3t) dt$. Find $F'(3)$.

Answer: E

By the Fundamental Theorem of Calculus $F'(x) = x^2 - 3x$, so $F'(3) = 9 - 9 = 0$.

Question 8 TRG-07 · Trigonometric functions and equations · spec MM4.6

Solve $\sin^2 x = \frac{3}{4}$ for $0 \leq x < 2\pi$. How many solutions are there?

Answer: E

$\sin x = \pm \frac{\sqrt{3}}{2}$; $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$. Four solutions.

Question 9 INT-16 · Integration · spec MM7.4

Given that $\int_0^4 f(x) dx = 10$, find $\int_0^4 (f(x) + 3x) dx - \int_4^0 f(x) dx$.

Answer: D

$$10 + 24 - (-10) = 44.$$

Question 10 SEQ-16 · Sequences and series · spec MM2.2

The sum of the first n terms of an arithmetic series is S_n . Given that $S_{10} = 145$ and $S_{20} = 490$, find the common difference.

Answer: A

$$S_{10} = 5(2a + 9d) = 145 \Rightarrow 2a + 9d = 29; S_{20} = 10(2a + 19d) = 490 \Rightarrow 2a + 19d = 49. \text{ Subtract: } 10d = 20, d = 2.$$

Question 11 BIN-13 · Binomial expansion · spec MM2.4

Use the binomial expansion to find the value of 1.01^5 correct to 4 decimal places.

Answer: B

$$(1 + 0.01)^5 = 1 + 5(0.01) + 10(0.0001) + 10(0.000001) + \dots = 1 + 0.05 + 0.001 + 0.00001 + \dots \approx 1.0510.$$

Question 12 DIF-09 · Differentiation · spec MM6.3

A rectangle has perimeter 40. What is its maximum possible area?

Answer: E

Sides x and $20 - x$: $A = 20x - x^2$, $A' = 20 - 2x = 0$ at $x = 10$, giving $A = 100$ (a square).

Question 13 IND-10 · Indices and surds · spec MM1.1

The value of $\frac{2^{n+2} + 2^n}{2^{n+1}}$ is

Answer: E

Factor 2^n : numerator $= 2^n(4 + 1) = 5 \cdot 2^n$; denominator $= 2 \cdot 2^n$. Quotient $= \frac{5}{2}$.

Question 14 ALG-13 · GCSE algebra · spec M4.5

Factorise $6x^2 - 7x - 20$.

Answer: C

$$(3x + 4)(2x - 5) = 6x^2 - 15x + 8x - 20 = 6x^2 - 7x - 20.$$

Question 15 PRO-05 · Probability · spec M7.2

A spinner has probability 0.15 of landing on red. It is spun 200 times. What is the expected number of times it lands on red?

Answer: A

$$0.15 \times 200 = 30.$$

Question 16 BIN-04 · Binomial expansion · spec MM2.4

What is the sum of the coefficients in the expansion of $(3x - 2)^7$?

Answer: D

$$\text{Put } x = 1: (3 - 2)^7 = 1.$$

Question 17 FUN-04 · Functions and transformations · spec MM1.7

Which of the following functions, each with domain the set of all real numbers, is one-to-one?

Answer: D

$x^3 + x$ is strictly increasing (its derivative $3x^2 + 1 > 0$), so different inputs give different outputs. $x^3 - x$ takes the value 0 at $x = -1, 0, 1$; the others are even or periodic.

Question 18 SIM-10 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $(x - 1)(x - 3)(x - 5) > 0$.

Answer: A

Sign chart: the product is positive for $x > 5$, negative for $3 < x < 5$, positive for $1 < x < 3$, negative for $x < 1$.

Question 19 TRI-10 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $\sin A : \sin B : \sin C = 3 : 5 : 7$. Find angle C .

Answer: A

$$\text{By the sine rule the sides are in ratio } 3 : 5 : 7. \cos C = \frac{9 + 25 - 49}{2 \cdot 3 \cdot 5} = -\frac{15}{30} = -\frac{1}{2}, \text{ so } C = 120^\circ.$$

Question 20 DIF-16 · Differentiation · spec MM6.3

The tangent to $y = x^3$ at the point (a, a^3) , where $a \neq 0$, meets the curve again at the point with x -coordinate

Answer: B

Tangent: $y = 3a^2x - 2a^3$. Solve $x^3 - 3a^2x + 2a^3 = 0$; $x = a$ is a double root, so $(x - a)^2(x + 2a) = 0$: the other intersection is $x = -2a$.